

Experiment 5

Aim - To Build the pipeline of jobs using Maven / Gradle / Ant in Jenkins, create a pipeline script to Test and deploy an application over the tomcat server

Programming in Jenkins:

Continuous Integration is a software development practice where members of a team integrate their

work frequently, usually each person integrates at least daily leading to multiple integrations per day.

Each integration is verified by an automated build (including test) to detect integration errors as

quickly as possible.” In simple way, Continuous integration (CI) is the practice of frequently building

and testing each change done to your code automatically.

Jenkins is a self-contained, open-source automation server which can be used to automate all sorts of

tasks related to building, testing, and delivering or deploying software.

Our first job will execute the shell commands. The freestyle project provides enough options and

features to build the complex jobs that you will need in your projects.

Example 1

Example 1.1: Deploying a freestyle app in Jenkins

Creating a job:

Naming the job and setting it as freestyle:

Selecting build type as “Execute shell”:

Entering a simple command for the shell execution:

Applying and saving the project configuration:

Building the project:

Console output (after building):

Example 1.2: Taking parameters through files

Contents of script example1.cmd:

Executing script example1.cmd on the terminal:

Modifying the Jenkins project to execute the script while supplying required parameters:

Console output after building the modified project:

Example 2

Example 2.1: Running a Java program under Jenkins

Creating a simple Java program:

Compiling and running the program on the terminal:

Creating a new freestyle project:

Configure new project:

Console output after building:

Example 3

Example 3.1: Parameterise build

Creating a new freestyle project:

Enabling parameterisation and adding a String parameter:

Configuring the string parameter as Fname:

Adding a choice parameter and configuring it as City with the following choices:

Creating a script which takes 2 arguments for name and city:

Configuring build steps:

Entering parameters for build:

Console output after building:

Example 3.2: Running a Java program with parameters

Creating a Java program with an input argument:

Testing the program on the terminal:

Creating a new freestyle project:

Parameterise the project by adding a string parameter as follows:

Configure the build steps:

Entering the parameter for the build:

Console output after building:

Example 5

Example 5.1: Running a Python program

Creating a simple Python script:

Running the Python script on the terminal:

Creating a new freestyle project:

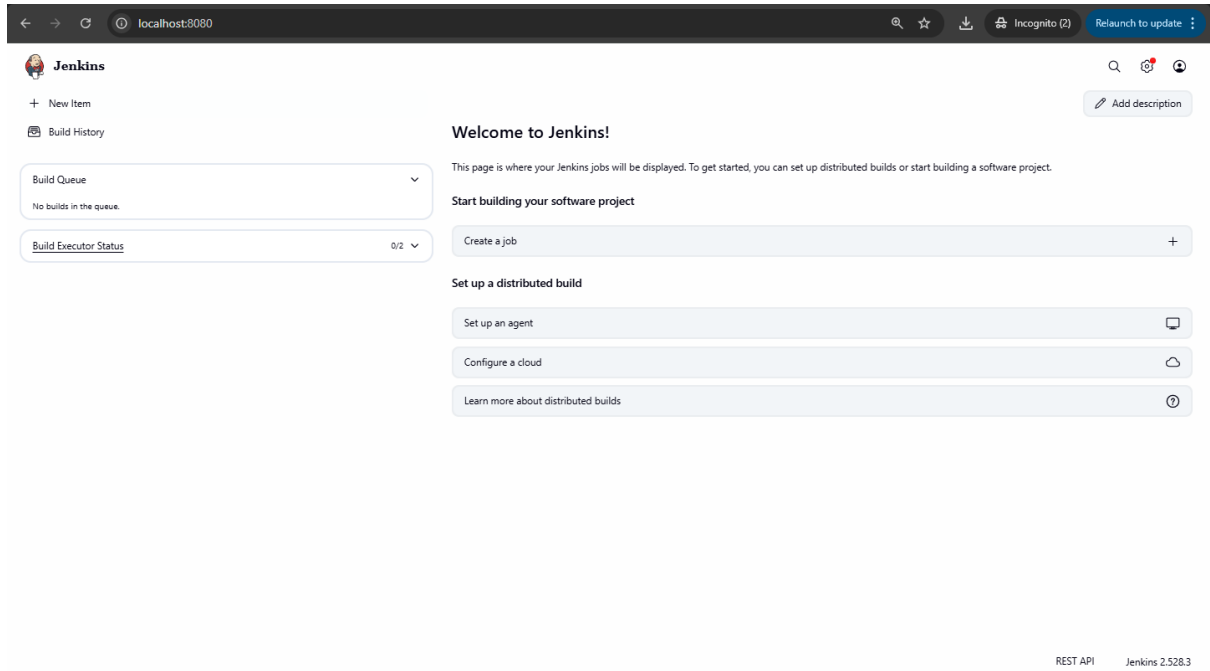
Parameterising the project with a string parameter as follows:

Configuring the build steps:

Setting the parameter for the build:

Console output after building:

Conclusion: Thus, we have successfully studied Continuous Integration and installed, configured, and understood programming with Jenkins.



The image shows the Jenkins 'Welcome to Jenkins!' page in a web browser. The browser's address bar shows 'localhost:8080'. The Jenkins logo is in the top left. On the left sidebar, there are links for 'New Item', 'Build History', 'Build Queue', and 'Build Executor Status'. The 'Build Queue' shows 'No builds in the queue.' and the 'Build Executor Status' shows '0/2'. The main content area has a 'Welcome to Jenkins!' heading, a brief description, and a 'Start building your software project' section with buttons for 'Create a job', 'Set up a distributed build' (which includes 'Set up an agent', 'Configure a cloud', and 'Learn more about distributed builds'), and 'Add description'. The bottom right corner shows 'REST API' and 'Jenkins 2.528.3'.

localhost:8080

Jenkins

+ New Item

Build History

Build Queue

No builds in the queue.

Build Executor Status

0/2

Welcome to Jenkins!

This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.

Start building your software project

Create a job

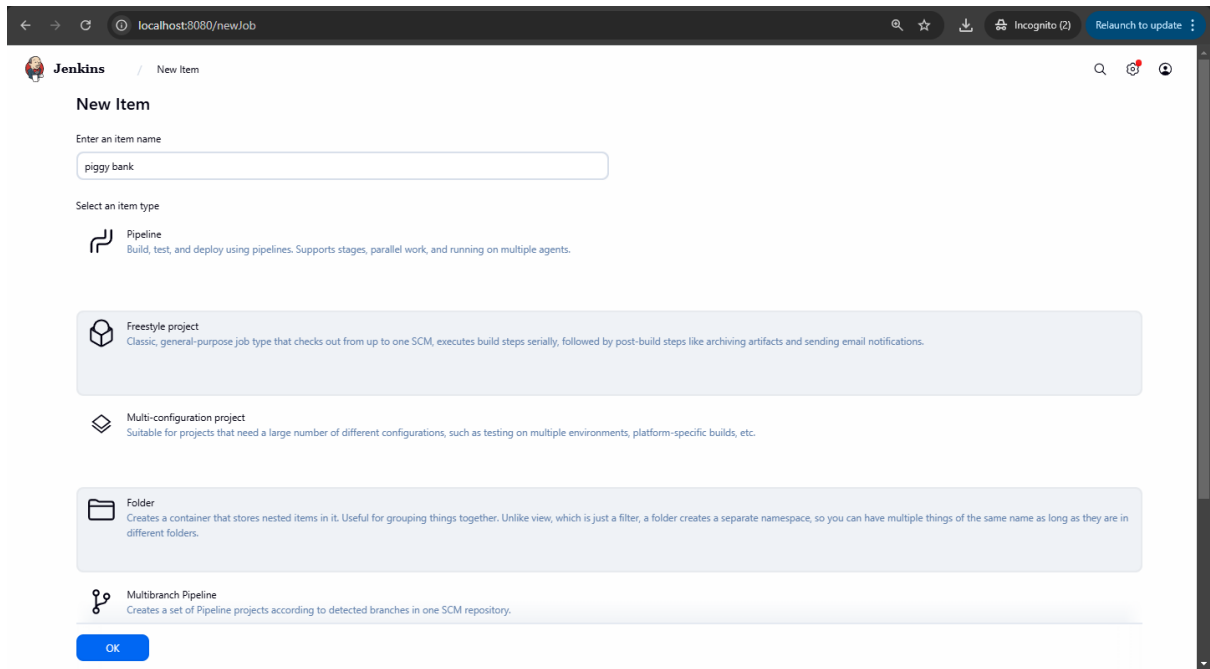
Set up a distributed build

Set up an agent

Configure a cloud

Learn more about distributed builds

REST API Jenkins 2.528.3



The image shows the Jenkins 'New Item' page in a web browser. The browser's address bar shows 'localhost:8080/newJob'. The Jenkins logo is in the top left. The page title is 'New Item'. There is a text input field for 'Enter an item name' with the value 'piggy bank'. Below it is a section 'Select an item type' with four options: 'Pipeline' (Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.), 'Freestyle project' (Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.), 'Multi-configuration project' (Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.), and 'Folder' (Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.). At the bottom is a 'Multibranch Pipeline' option (Creates a set of Pipeline projects according to detected branches in one SCM repository.). An 'OK' button is at the bottom left.

localhost:8080/newJob

Jenkins / New Item

New Item

Enter an item name

piggy bank

Select an item type

Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.

Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

Folder

Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.

Multibranch Pipeline

Creates a set of Pipeline projects according to detected branches in one SCM repository.

OK

The image displays two screenshots of the Jenkins web interface, showing the configuration page for a job named "piggy bank".

Top Screenshot (General Tab):

- The browser address bar shows `localhost:8080/job/piggy%20bank/configure`.
- The left sidebar shows the "Configure" menu with options: General (selected), Source Code Management, Triggers, Environment, Build Steps, and Post-build Actions.
- The main content area is titled "General" and includes a description field (empty).
- Below the description, there are several checkboxes with help icons (question marks):
 - ☐ Discard old builds
 - ☐ GitHub project
 - ☐ This project is parameterized
 - ☐ Throttle builds
 - ☐ Execute concurrent builds if necessary
- An "Advanced" dropdown menu is visible.
- The "Source Code Management" section shows "None" selected for the repository type.
- Buttons for "Save" and "Apply" are at the bottom.
- A status bar at the bottom right indicates "REST API" and "Jenkins 2.528.3".

Bottom Screenshot (Build Steps Tab):

- The browser address bar shows `localhost:8080/job/piggy%20bank/configure`.
- The left sidebar shows the "Configure" menu with options: General, Source Code Management, Triggers, Environment, Build Steps (selected), and Post-build Actions.
- The main content area is titled "Build Steps" and includes a "Command" field with the text `echo "Hello Tsec"`.
- Below the command field, there is an "Advanced" dropdown menu.
- A button for "+ Add build step" is visible.
- The "Post-build Actions" section shows a button for "+ Add post-build action".
- Buttons for "Save" and "Apply" are at the bottom.
- A green notification bar at the bottom left says "Saved".
- A status bar at the bottom right indicates "REST API" and "Jenkins 2.528.3".

The screenshot shows the Jenkins Configuration page for a job named 'piggy bank'. The left sidebar contains a 'Configure' section with links to General, Source Code Management, Triggers, Environment (selected), Build Steps, and Post-build Actions. The main area is divided into two sections: 'Configure' and 'Build Steps'. The 'Configure' section has several checkboxes: 'Delete workspace before build starts', 'Use secret text(s) or file(s)', 'Add timestamps to the Console Output', 'Inspect build log for published build scans', 'Terminate a build if it's stuck', and 'With Ant'. The 'Build Steps' section has a title 'Build Steps' and a description 'Automate your build process with ordered tasks like code compilation, testing, and deployment.' Below this is a dashed box containing a step named 'Execute Windows batch command'. The 'Command' field contains the text 'echo Hello Tisha'. A green 'Saved' message is visible at the bottom left, and 'Save' and 'Apply' buttons are at the bottom right.

localhost:8080/job/piggy%20bank/configure

Jenkins / piggy bank / Configuration

Configure

- General
- Source Code Management
- Triggers
- Environment**
- Build Steps
- Post-build Actions

☐ Delete workspace before build starts

☐ Use secret text(s) or file(s) ?

☐ Add timestamps to the Console Output

☐ Inspect build log for published build scans

☐ Terminate a build if it's stuck

☐ With Ant ?

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute Windows batch command ?

Command

See the list of available environment variables

echo Hello Tisha

Advanced

Save Apply

✓ Saved

The screenshot shows the Jenkins Job Overview page for a job named 'piggy bank'. The left sidebar contains a 'Status' section with links to Changes, Workspace, Build Now, Configure, Delete Project, and Rename. The main area is divided into two sections: 'Status' and 'Permalinks'. The 'Status' section has a green checkmark and the text 'piggy bank'. The 'Permalinks' section contains a list of build links: 'Last build (#13), 2 sec ago', 'Last stable build (#13), 2 sec ago', 'Last successful build (#13), 2 sec ago', 'Last failed build (#12), 1 min 32 sec ago', 'Last unsuccessful build (#12), 1 min 32 sec ago', and 'Last completed build (#13), 2 sec ago'. Below this is a 'Builds' section with a search bar and a list of builds. The list shows builds #1 through #13, with build #13 being the most recent and successful. The bottom of the page shows the URL 'localhost:8080/job/piggy bank/ws/'.

localhost:8080/job/piggy%20bank/

Jenkins / piggy bank

Status

Changes

Workspace

Build Now

Configure

Delete Project

Rename

✓ piggy bank

Add description

Permalinks

- Last build (#13), 2 sec ago
- Last stable build (#13), 2 sec ago
- Last successful build (#13), 2 sec ago
- Last failed build (#12), 1 min 32 sec ago
- Last unsuccessful build (#12), 1 min 32 sec ago
- Last completed build (#13), 2 sec ago

Builds

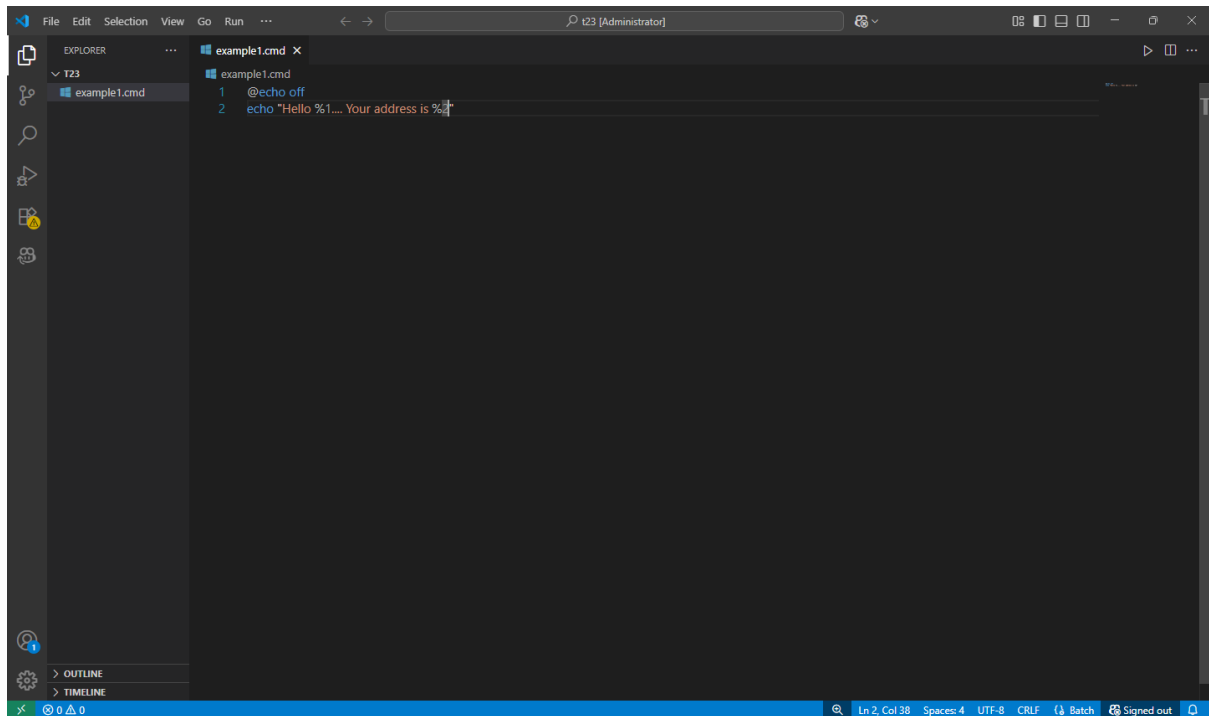
Filter

Today

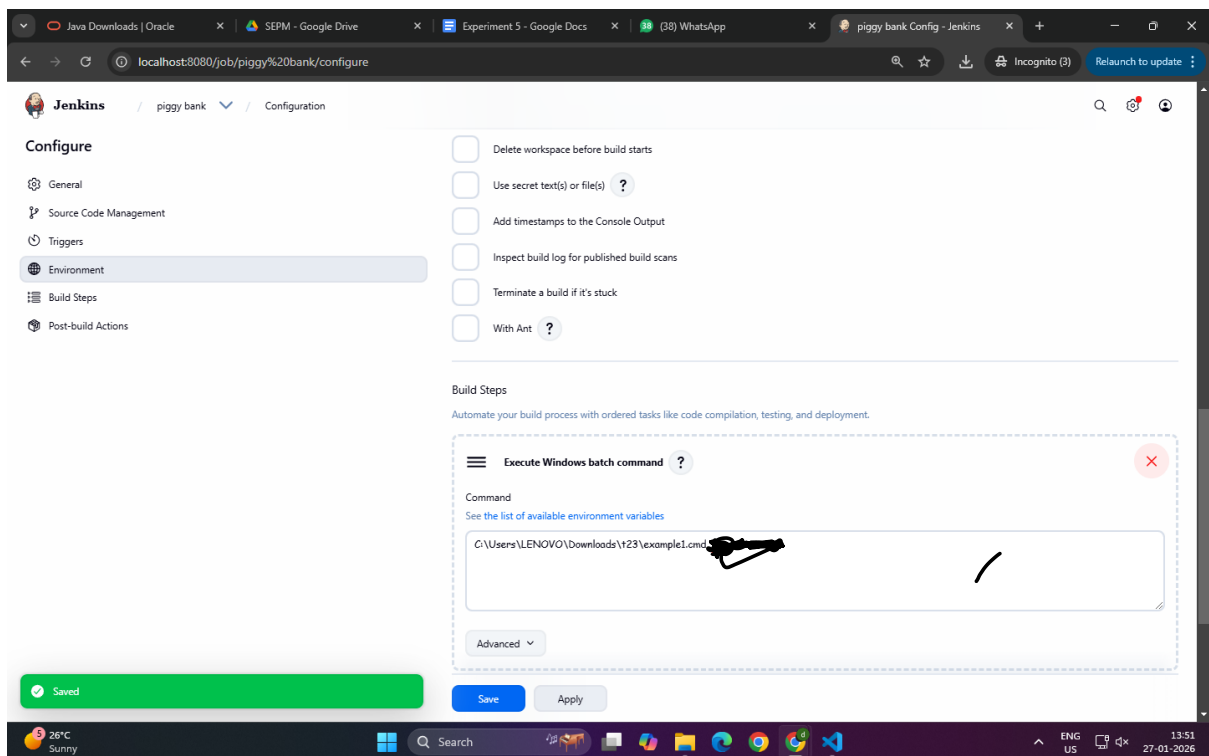
- ✓ #13 1:45 PM
- ✗ #12 1:43 PM
- ✗ #11 1:42 PM
- ✗ #10 1:42 PM
- ✗ #9 1:41 PM
- ✗ #8 1:40 PM
- ✗ #7 1:40 PM

localhost:8080/job/piggy bank/ws/

1.2



```
example1.cmd
1 @echo off
2 echo "Hello %1... Your address is %~p"
```



Example 2

The image displays two screenshots related to a Java project setup.

The top screenshot shows the Jenkins configuration page for a job named "Example 2". The "General" tab is selected, and the "Enabled" toggle is turned on. The "Description" field is empty. Below the "Description" field, there are several checkboxes: "Discard old builds", "GitHub project", "This project is parameterized", "Throttle builds", and "Execute concurrent builds if necessary". The "Advanced" section is collapsed. The "Source Code Management" section is also collapsed, showing "None" as the selected option. The "Save" and "Apply" buttons are visible at the bottom.

The bottom screenshot shows the Visual Studio Code editor. The Explorer pane on the left shows the project structure: "T23" (a folder) containing "example1.cmd", "Example2.class", "Example2.java", and "Main.class". The "Example2.java" file is open in the editor, showing the following code:

```
1 class Example2 {  
2     public static void main(String args[]) {  
3         System.out.println("Hello World!");  
4     }  
5 }
```

The Output pane at the bottom shows the command prompt output: "(base) PS C:\Users\LENOVO\Downloads\t23>". The status bar at the bottom indicates "Ln 2, Col 41", "Spaces: 4", "UTF-8", "CRLF", "Java", and "Signed out".

The image shows two screenshots of the Jenkins web interface. The top screenshot displays the 'Configure' page for a job named 'Example 2'. The left sidebar shows the configuration menu with options: General, Source Code Management, Triggers, Environment (selected), Build Steps, and Post-build Actions. The main area has several checkboxes for configuration: 'Delete workspace before build starts', 'Use secret text(s) or file(s)', 'Add timestamps to the Console Output', 'Inspect build log for published build scans', 'Terminate a build if it's stuck', and 'With Ant'. Below these is the 'Build Steps' section, which includes a step named 'Execute Windows batch command' with a command field containing 'C:\Users\LENOVO\Downloads\t23\Example2.java'. The bottom screenshot shows the 'Example 2' job page. The left sidebar has options: Status (selected), Changes, Workspace, Build Now, Configure, Delete Project, and Rename. The main area shows 'Permalinks' and a 'Builds' section with a single build entry: '#1 2:00 PM'. A green notification bar at the bottom states 'Build scheduled'. The bottom status bar shows 'localhost:8080/job/Example 2/build?delay=0sec', 'REST API', 'Jenkins 2.528.3', and system information like 'Nifty bank' and '27-01-2026'.

localhost:8080/job/Example%202/configure

Jenkins / Example 2 / Configuration

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute Windows batch command

Command

See the list of available environment variables

C:\Users\LENOVO\Downloads\t23\Example2.java

Advanced

Save Apply

localhost:8080/job/Example%202/

Jenkins / Example 2

Status

Changes

Workspace

Build Now

Configure

Delete Project

Rename

Example 2

Permalinks

Builds

Today

#1 2:00 PM

Build scheduled

localhost:8080/job/Example 2/build?delay=0sec

REST API Jenkins 2.528.3

Nifty bank

27-01-2026

Example 3

The image shows two screenshots of the Jenkins web interface. The top screenshot is the 'New Item' page, where a new job named 'Example 3' is being created. The 'Pipeline' option is selected. The bottom screenshot is the 'Configure' page for 'Example 3'. The 'General' tab is active, and the 'This project is parameterized' checkbox is checked. A dropdown menu is open, showing various parameter types: Boolean Parameter, Choice Parameter, Credentials Parameter, File Parameter, Multi-line String Parameter, Password Parameter, Run Parameter, and String Parameter. The 'String Parameter' option is highlighted. The 'Save' button is visible at the bottom.

New Item

Enter an item name
Example 3

Select an item type

- Pipeline**
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
- Freestyle project**
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
- Multi-configuration project**
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
- Folder**
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.
- Multibranch Pipeline**
Creates a set of Pipeline projects according to detected branches in one SCM repository.

Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

☐ GitHub project

☒ This project is parameterized ?

+ Add Parameter

Filter

- Boolean Parameter
- Choice Parameter
- Credentials Parameter
- File Parameter
- Multi-line String Parameter
- Password Parameter
- Run Parameter
- String Parameter

☐ Trigger builds remotely (e.g., from scripts) ?

☐ Build after other projects are built ?

Save Apply

The image displays two screenshots of the Jenkins configuration page for a job named 'Example 3'. The top screenshot shows the 'String Parameter' configuration, and the bottom screenshot shows the 'Choice Parameter' configuration.

Top Screenshot: String Parameter Configuration

- Configuration Page:** The page is titled 'Configure' and shows a sidebar with options: General, Source Code Management, Triggers, Environment, Build Steps, and Post-build Actions. The 'General' tab is selected.
- Parameter Type:** A blue checkmark indicates 'This project is parameterized'.
- String Parameter:** A new parameter is being added with the following fields:
 - Name:** fname
 - Default Value:** (empty)
 - Description:** (empty)
 - Trim the string:** (unchecked)
- Buttons:** 'Save' and 'Apply' buttons are at the bottom.

Bottom Screenshot: Choice Parameter Configuration

- Configuration Page:** The page is titled 'Configure' and shows the same sidebar as the top screenshot.
- Parameter Type:** A blue checkmark indicates 'This project is parameterized'.
- Choice Parameter:** A new parameter is being added with the following fields:
 - Name:** City
 - Choices:** Bandra, Kalyan, Mulund
 - Description:** (empty)
- Buttons:** 'Save' and 'Apply' buttons are at the bottom.

The image shows a Jenkins configuration page for a job named 'Example 3' and a Visual Studio Code editor window.

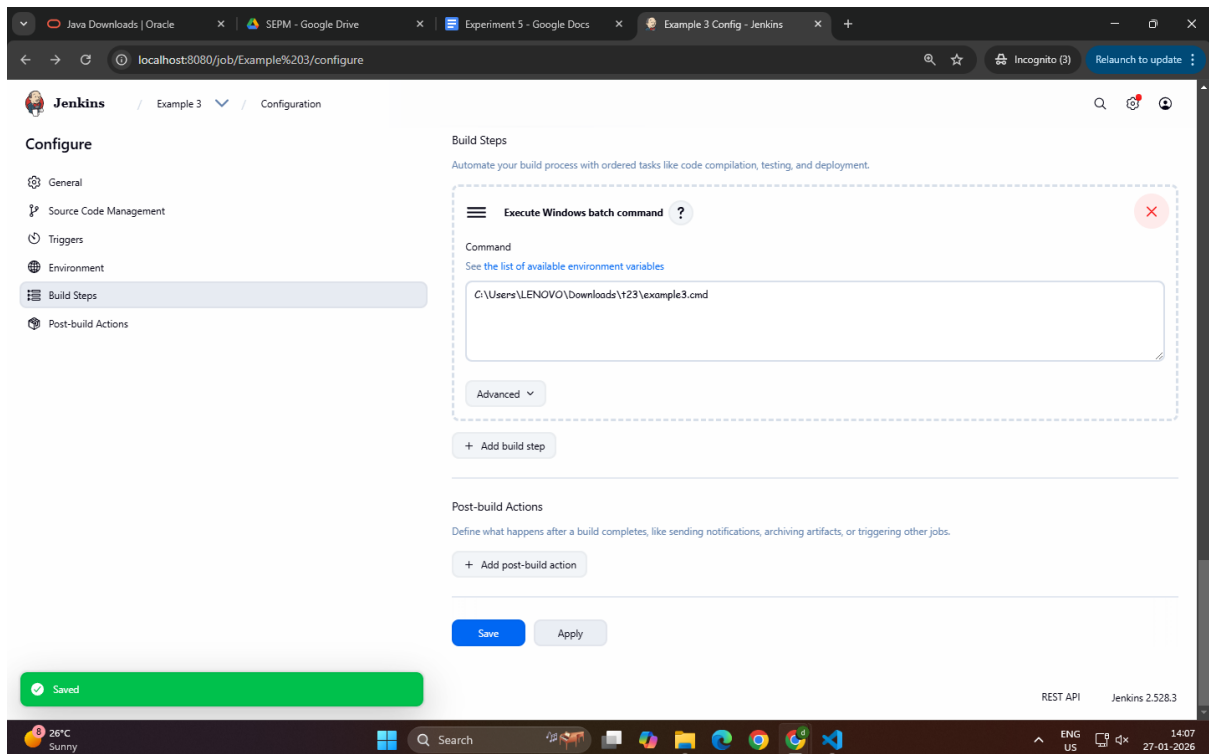
Jenkins Configuration Page:

- URL: `localhost:8080/job/Example%203/configure`
- Page Title: Jenkins / Example 3 / Configuration
- Left sidebar: Configuration options (General, Source Code Management, Triggers, Environment, Build Steps, Post-build Actions).
- Main content: 'This project is parameterized' checkbox is checked. Below it, two parameter types are visible: 'String Parameter' and 'Choice Parameter'. The 'String Parameter' form has fields for Name (fname), Default Value, and Description. The 'Choice Parameter' form has a Name field.
- Buttons: 'Save' and 'Apply'.
- Notification: A green 'Saved' message is displayed.

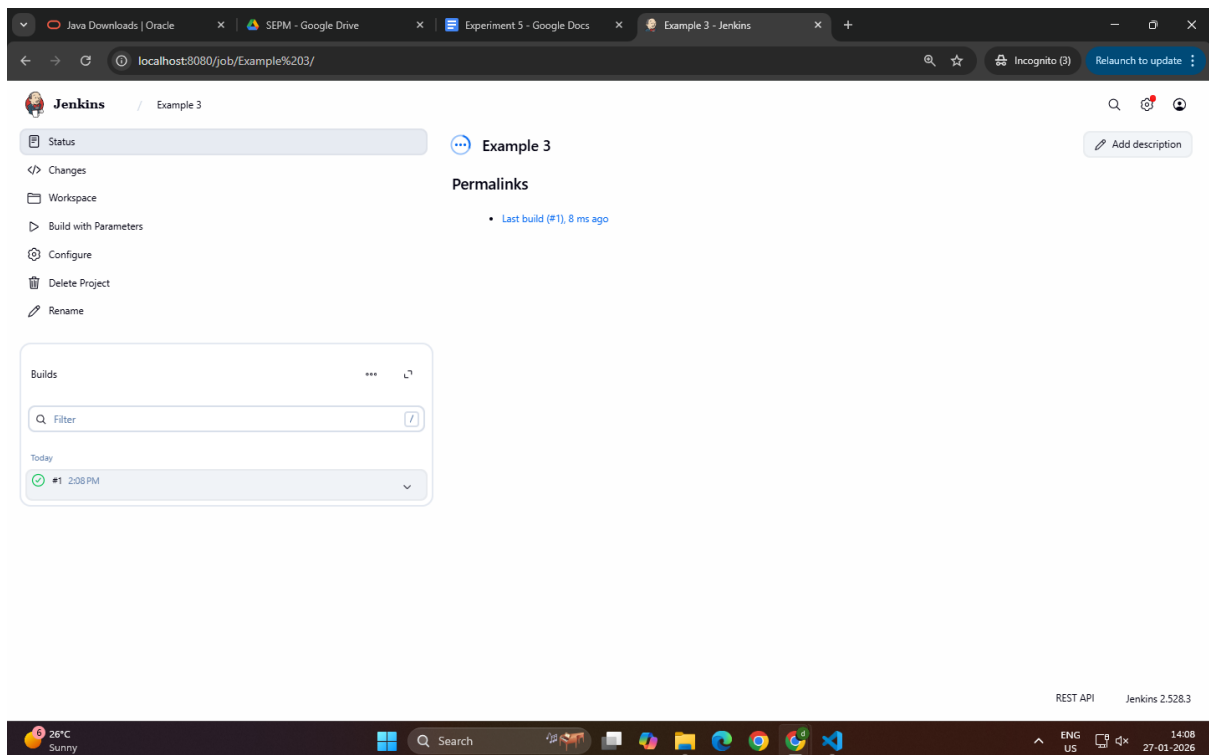
Visual Studio Code Editor:

- File Explorer: Shows a project structure with files `example1.cmd`, `Example2.class`, `Example2.java`, `example3.cmd`, and `Main.class`.
- Editor: The file `example3.cmd` is open, showing the following content:

```
1 @echo off
2 echo "Hello %1.... Your city is %2"
```
- Terminal: A PowerShell terminal window is open, showing the command prompt `(base) PS C:\Users\LENOVO\Downloads\t23>`.
- Status Bar: Shows 'Ln 2, Col 29', 'Spaces: 4', 'UTF-8', 'CRLF', 'Batch', and 'Signed out'.

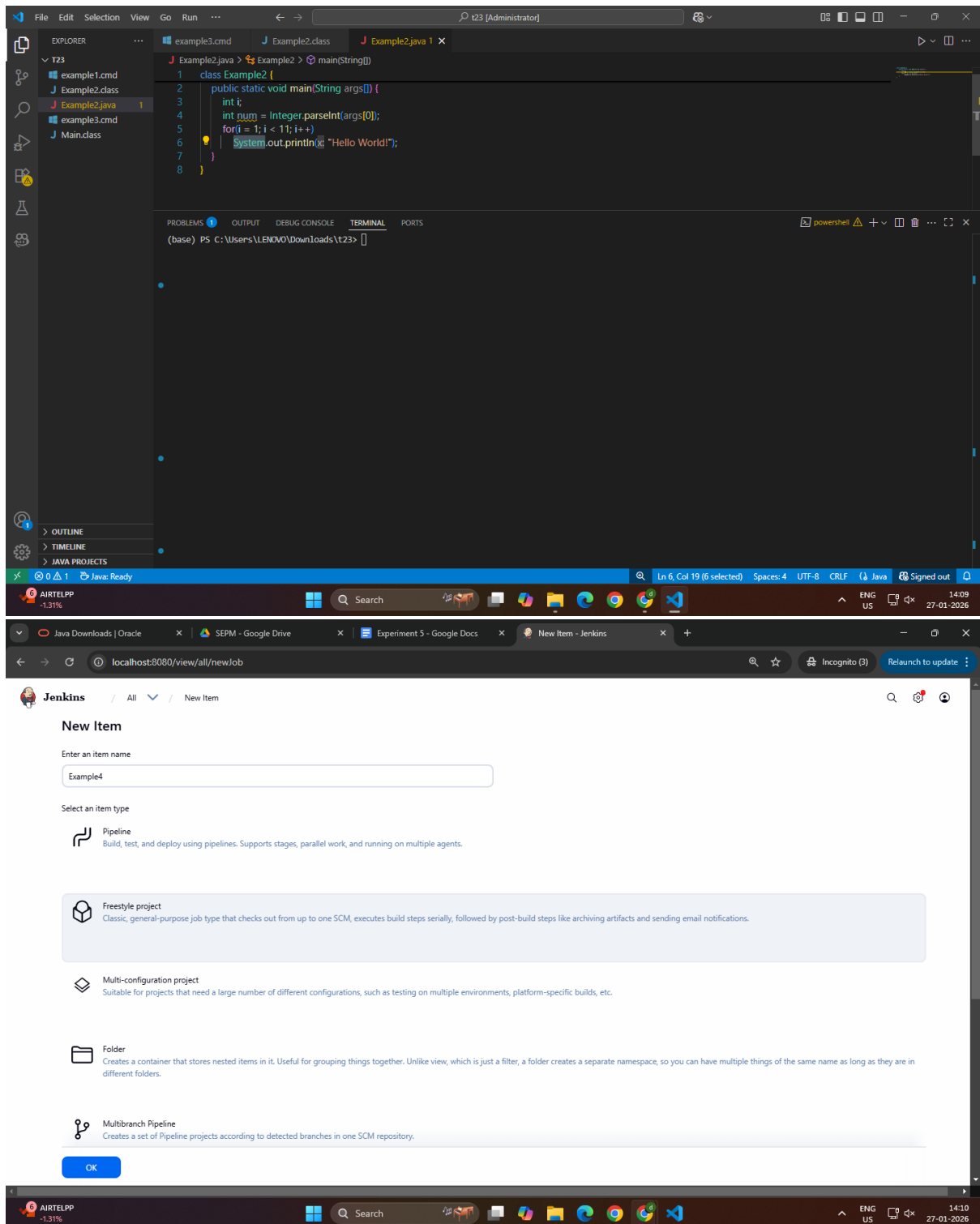


The screenshot shows the Jenkins Configuration page for a job named 'Example 3'. The left sidebar contains a 'Configure' section with a list of tabs: General, Source Code Management, Triggers, Environment, Build Steps, and Post-build Actions. The 'Build Steps' tab is selected. The main area is titled 'Build Steps' and contains a section for 'Execute Windows batch command'. The command field is populated with 'C:\Users\LENOVO\Downloads\t23\example3.cmd'. Below the command field is an 'Advanced' dropdown menu. At the bottom of the configuration area are 'Save' and 'Apply' buttons. A green 'Saved' notification bar is visible at the bottom left. The bottom status bar shows 'REST API' and 'Jenkins 2.528.3'.



The screenshot shows the Jenkins Job Overview page for 'Example 3'. The left sidebar contains a 'Status' section with a list of tabs: Changes, Workspace, Build with Parameters, Configure, Delete Project, and Rename. The 'Configure' tab is selected. The main area is titled 'Example 3' and contains a 'Permalinks' section with a link to 'Last build (#1), 8 ms ago'. Below the permalinks is a 'Builds' section with a search filter and a list of builds. The first build is '#1' with a status of 'Completed' and a time of '2:08 PM'. The bottom status bar shows 'REST API' and 'Jenkins 2.528.3'.

3.2



The image displays two screenshots of the Jenkins web interface, showing the configuration for a job named 'Example4'.

Top Screenshot: String Parameter Configuration

- The browser address bar shows `localhost:8080/job/Example4/configure`.
- The Jenkins configuration page for 'Example4' is shown, with the 'General' tab selected.
- The 'This project is parameterized' checkbox is checked.
- A 'String Parameter' is being configured with the name 'num'.
- The 'Default Value' field is empty.
- The 'Description' field is empty.
- The 'Trim the string' checkbox is unchecked.
- The 'Add Parameter' button is visible at the bottom.

Bottom Screenshot: Build Steps Configuration

- The browser address bar shows `localhost:8080/job/Example4/configure`.
- The Jenkins configuration page for 'Example4' is shown, with the 'Build Steps' tab selected.
- The 'Build Steps' section is titled 'Automate your build process with ordered tasks like code compilation, testing, and deployment.'
- An 'Execute Windows batch command' step is configured with the command `C:\Users\LENOVO\Downloads\T23\Example4.java %num%`.
- The 'Advanced' dropdown is visible.
- The 'Add build step' button is visible.
- The 'Post-build Actions' section is titled 'Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.'
- The 'Add post-build action' button is visible.
- The 'Save' and 'Apply' buttons are at the bottom.

A green 'Saved' notification bar is visible at the bottom of the second screenshot.

The screenshot displays the Jenkins web interface in a browser window. The address bar shows the URL `localhost:8080/job/Example4/build?delay=0sec`. The Jenkins logo and the job name 'Example4' are visible at the top left. The main heading is 'Project Example4'. Below it, a message states 'This build requires parameters:'. A parameter named 'num' is shown with a text input field containing the value '4'. Below the input field are two buttons: 'Build' (green) and 'Cancel' (grey). On the left sidebar, there are links for 'Status', 'Changes', 'Workspace', 'Build with Parameters', 'Configure', 'Delete Project', and 'Rename'. At the bottom left, a 'Builds' section shows a list of builds. The first build is labeled '#1' and occurred at '2:12 PM'. The bottom of the image shows a Windows taskbar with various application icons and system information, including the date '27-01-2026' and time '14:12'.